

Getting Spectra from PlasMol

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1 Parallel orientation and the three recorded series

This PlasMol calculation is hybrid NP + molecule in the parallel geometry. A 25nm Au sphere sits at the origin. The Na atom is on the $+x$ axis at $d = 26.451\text{nm}$ (gap 1.451nm). The source is a transverse plane wave arranged so that the driving field at the atom is along the line from the nanoparticle center to the atom (Figure 1). RT-TDDFT (LC- ω PBE / 6-31G*, IP-tuned range-separation $\omega = 0.339181$, static CAP) is advanced under the length-gauge coupling to the Meep field sampled at the atomic site.

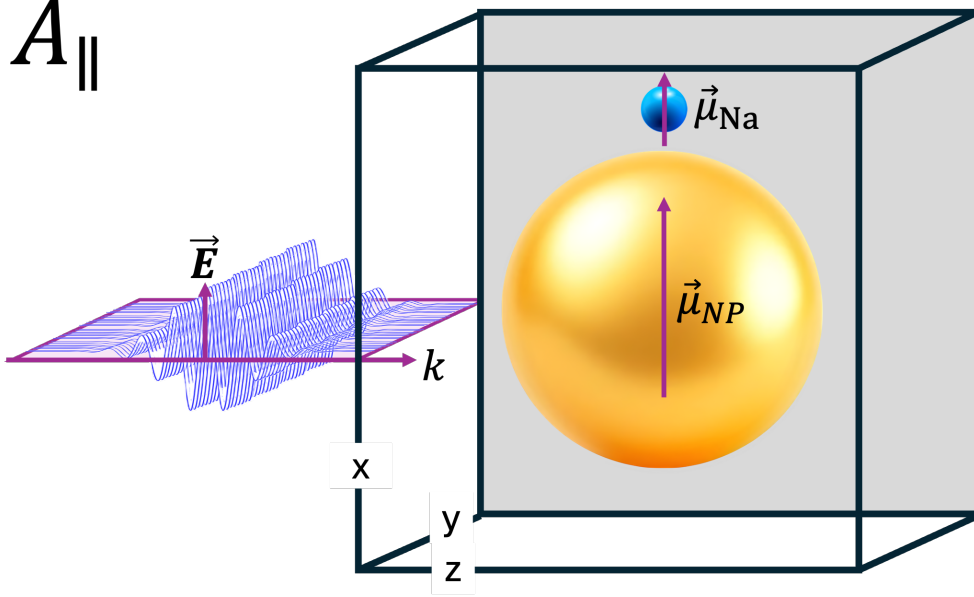


Figure 1: Parallel (radial) orientation, from the PlasMol documentation. The incident wave is transverse ($\mathbf{k} \perp \mathbf{E}$). Both μ_{NP} and μ_{Na} lie along the nanoparticle–molecule axis $\hat{\mathbf{r}}$. In the production job the atom is on the $+x$ axis at $d = 26.45\text{nm}$ (gap 1.45nm). Time traces at the Na site are in Figure 2.

Every production run writes three real functions of time at the molecular location:

- $E_{loc}(t)$: the electric field felt by the molecule (nanoparticle scattering included).
- $\mu(t)$: the induced dipole of the *molecule* (RT-TDDFT).
- $E_{inc}(t)$: the electric field at the same sample point in a vacuum reference run (same source, no nanoparticle, no molecule).

In the frequency domain the local field is the incident wave plus whatever the gold sphere (and, weakly, the molecule reflecting off the sphere) scatter,

$$E_{loc}(\omega) = E_{inc}(\omega) + E_{scat}(\omega). \quad (1)$$

The three hybrid traces are shown in Figure 2.

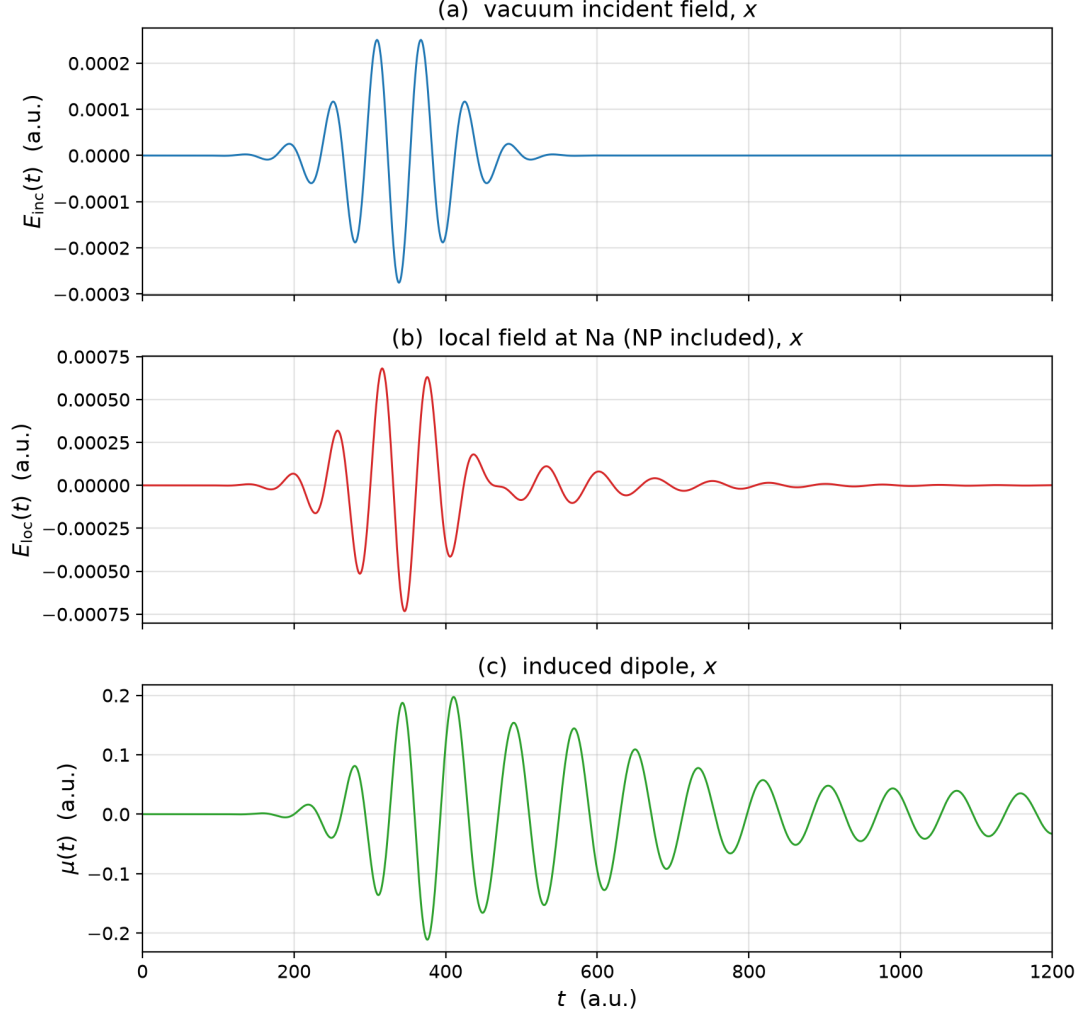


Figure 2: Driven (x) components through the arrival of the Meep pulse. (a) $E_{\text{inc}}(t)$. (b) $E_{\text{loc}}(t)$, larger and ringing after the vacuum pulse has gone. (c) $\mu(t)$, which continues to oscillate after both fields have died until about $t = 4000$ a.u..

2 Fourier transforms and linear response

2.1 Beginning equations

Frequency domain spectra are formed with the same discrete transform PlasMol uses,

$$f(\omega_k) = \Delta t \sum_{n=0}^{N-1} f(t_n) e^{-i\omega_k t_n}, \quad \omega_k = \frac{2\pi k}{N\Delta t}. \quad (2)$$

Even though the three traces collected are real, their transforms are nevertheless complex (Figure 3).

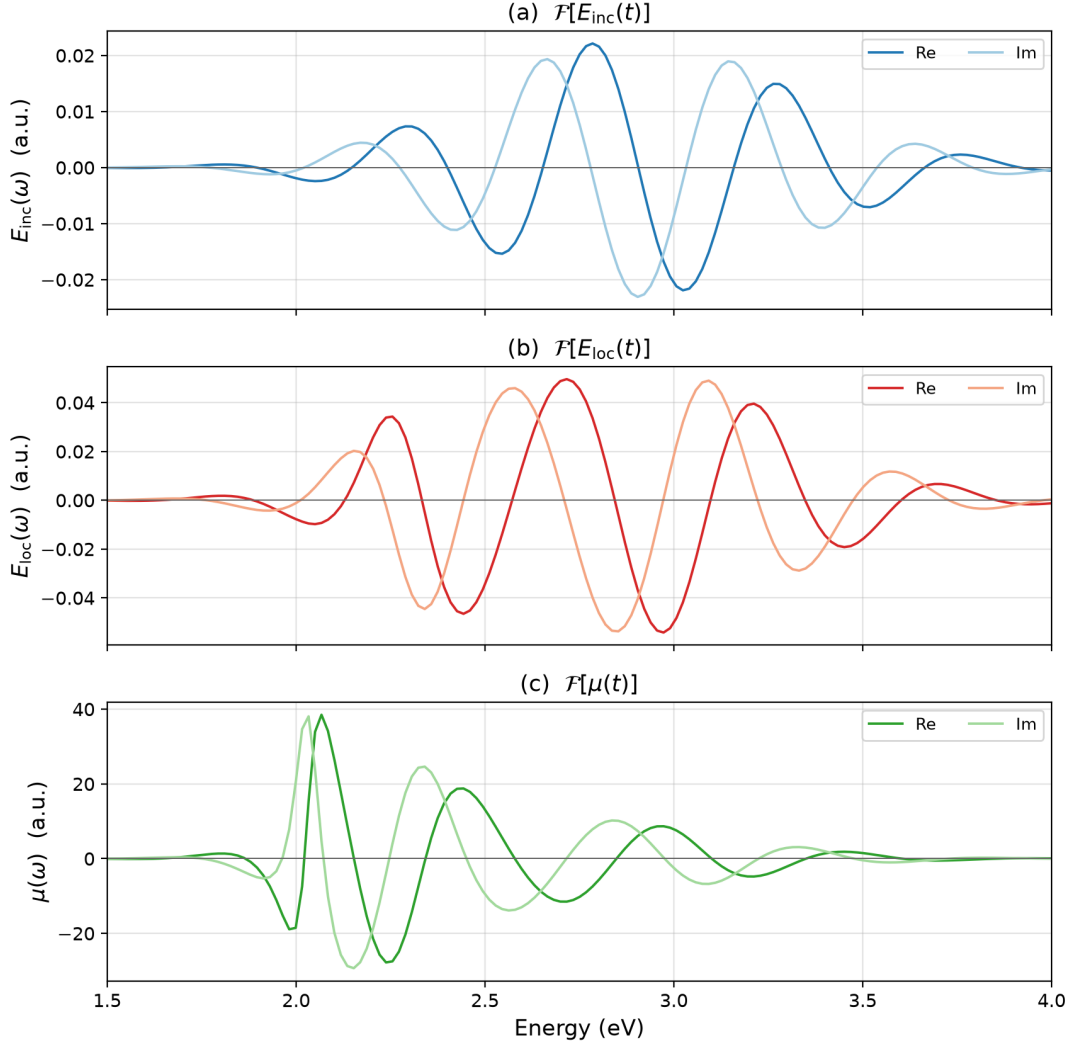


Figure 3: Real and imaginary parts of $E_{\text{inc}}(\omega)$, $E_{\text{loc}}(\omega)$, and $\mu(\omega)$. The DFT oscillates because the time-dependent fields are not centered at $t = 0$.

Taking the magnitude of the DFTs in Figure 3 gives what is traditionally viewed as an energy spectrum. Figure 4 is that construction for the incident field: $|E_{\text{inc}}(\omega)|$ is the envelope of the Meep pulse, peaked near 2.84 eV.

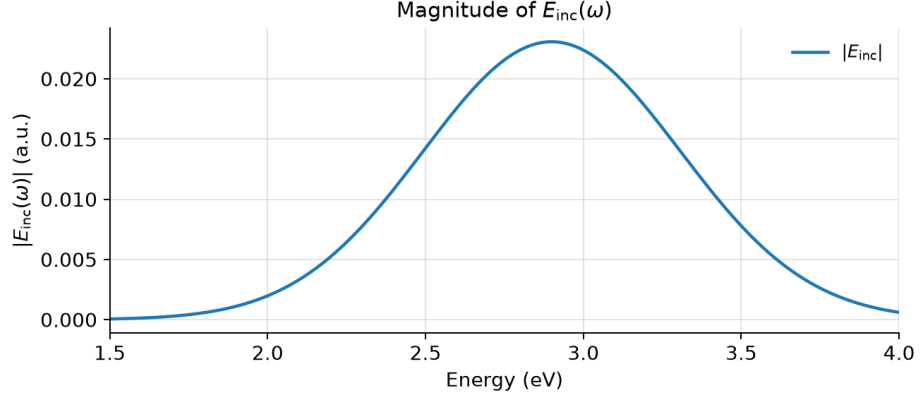


Figure 4: Magnitude of $E_{\text{inc}}(\omega)$ from the vacuum reference. This is the usual energy spectrum of the incident pulse; the delay phase that oscillates in Figure 3 does not appear in $|E_{\text{inc}}|$.

If the simulation were centered around $t = 0$, the Fourier transform would not have the pulsing seen in Figure 3. Figure 5 is an illustrative example, not a production hybrid run: the same vacuum E_{inc} with its peak shifted to the origin. The real part is then a smooth envelope and the imaginary part is nearly gone; $|E_{\text{inc}}(\omega)|$ is unchanged by the shift.

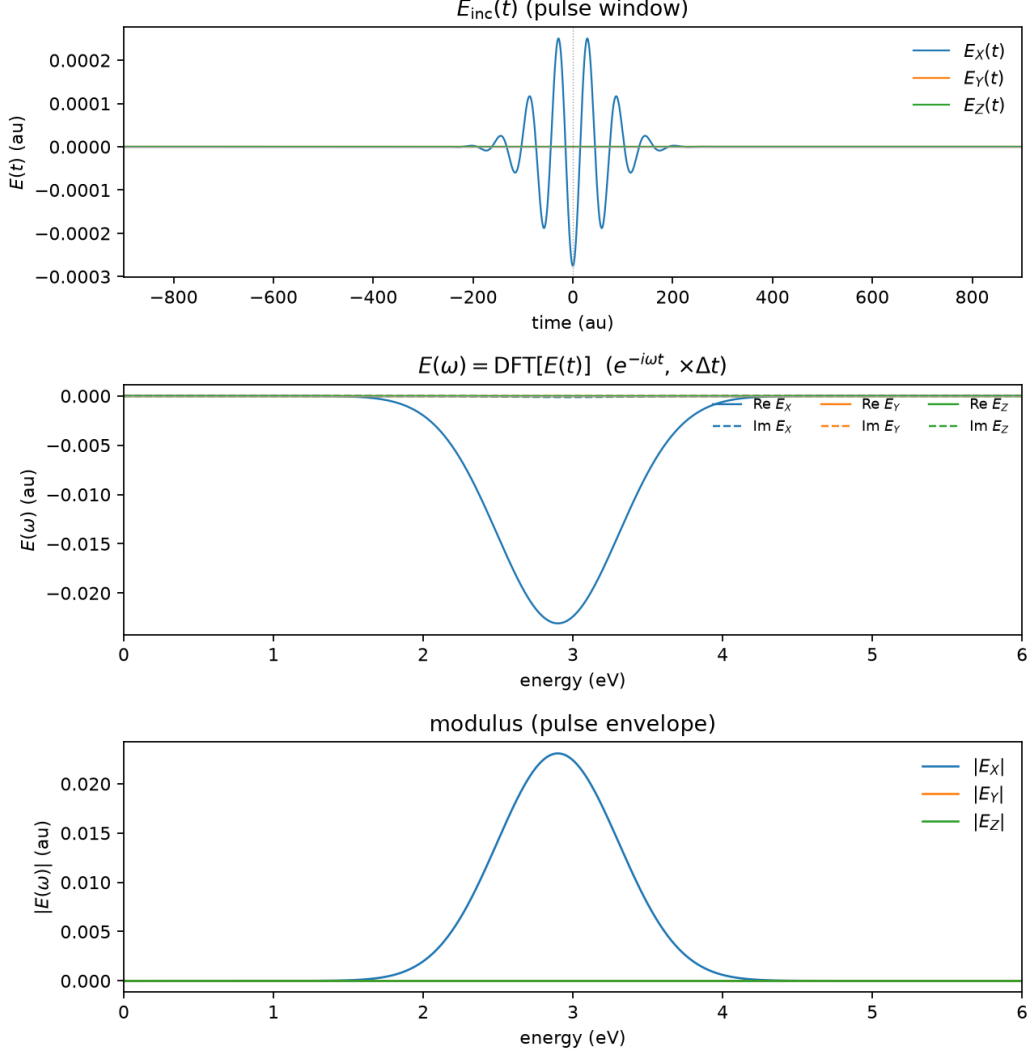


Figure 5: Illustrative example. The vacuum incident pulse of Figure 2 recentered at $t = 0$. Top: time trace. Middle: real and imaginary parts of the DFT; the delay pulsing of Figure 3 is absent. Bottom: $|E_{\text{inc}}(\omega)|$, the same envelope as Figure 4.

3 Power of the molecule

3.1 Instantaneous power

The instantaneous power delivered to a classical dipole by the electric field at its location is

$$P(t) = \dot{\mu}(t)E_{\text{loc}}(t). \quad (3)$$

This is the rate at which the local field does work on the oscillating charge distribution. Ideally, the energy that enters (3) would leave the electromagnetic field and sit in the atom or be subsequently removed by the CAP (though that is not technically the case within the discrete nature of PlasMol).

For future steps, it is useful to write a real field in phasor notation,

$$E(t) = \text{Re}[\tilde{E}e^{-i\omega t}], \quad \mu(t) = \text{Re}[\tilde{\mu}e^{-i\omega t}], \quad (4)$$

with independent phasors

$$\tilde{E} = |E|e^{i\varphi_E}, \quad \tilde{\mu} = |\mu|e^{i\varphi_\mu}. \quad (5)$$

Linear response of the molecule is always to the field at its location,

$$\mu(\omega) = \alpha_m(\omega) E_{\text{loc}}(\omega), \quad (6)$$

or, in the same phasors, $\tilde{\mu} = \alpha_m \tilde{E}_{\text{loc}}$. Here α_m is the molecular polarizability of the RT-TDDFT model, CAP included. The two phases are not assumed equal: if they were, $\mu(t)$ would be in phase with $E(t)$ and the cycle-averaged power would vanish. The physical field is always the real part. Under (4) the time derivative is

$$\dot{\mu}(t) = \text{Re}[-i\omega \tilde{\mu} e^{-i\omega t}]. \quad (7)$$

3.2 Cycle-averaged power

Instantaneous power oscillates. Half the cycle the field does work on the dipole; the other half the dipole does work on the field. That sloshing is reactive energy, stored and returned every period. It is not absorption. In order to get absorption, we want to see the net transfer of power, leftover after many cycles:

$$\begin{aligned} \bar{P} &= \frac{1}{T} \int_0^T P(t) dt \\ &= \frac{1}{T} \int_0^T \dot{\mu}(t) E_{\text{loc}}(t) dt \\ &= \frac{1}{T} \int_0^T \text{Re}[-i\omega \tilde{\mu} e^{-i\omega t}] \text{Re}[\tilde{E}_{\text{loc}} e^{-i\omega t}] dt \end{aligned}$$

Write $A = \text{Re}[\tilde{A} e^{-i\omega t}]$ and $B = \text{Re}[\tilde{B} e^{-i\omega t}]$. Without loss of generality,

$$A = \text{Re}[\tilde{A} e^{-i\omega t}] = \frac{1}{2} (\tilde{A} e^{-i\omega t} + \tilde{A}^* e^{i\omega t}), \quad (8)$$

so the product is

$$AB = \frac{1}{4} (\tilde{A} \tilde{B} e^{-2i\omega t} + \tilde{A} \tilde{B}^* + \tilde{A}^* \tilde{B} + \tilde{A}^* \tilde{B}^* e^{2i\omega t}). \quad (9)$$

The double-frequency terms average to zero over one period $T = 2\pi/\omega$ and both $\tilde{A}$ and $\tilde{B}$ are time independent. The cycle average is therefore

$$\overline{AB} = \frac{1}{T} \int_0^T AB dt = \frac{1}{T} \left(\frac{1}{4} (\tilde{A} \tilde{B}^* + \tilde{A}^* \tilde{B}) \right) T = \frac{1}{2} \text{Re}[\tilde{A} \tilde{B}^*] \quad (10)$$

where the following identity was used:

$$CD^* + C^*D = CD^* + (CD^*)^* = 2 \text{Re}[CD^*]. \quad (11)$$

Identify $A = \text{Re}[-i\omega \tilde{\mu} e^{-i\omega t}] \implies \tilde{A} = -i\omega \tilde{\mu}$ and $B = \text{Re}[\tilde{E}_{\text{loc}} e^{-i\omega t}] \implies \tilde{B} = \tilde{E}_{\text{loc}}$. Then

$$\bar{P} = \frac{1}{2} \text{Re}[(-i\omega \tilde{\mu}) \tilde{E}_{\text{loc}}^*]. \quad (12)$$

For real ω one has $\text{Re}[-iz] = \text{Im}[z]$, and (12) becomes

$$\bar{P} = \frac{\omega}{2} \text{Im}[\tilde{\mu} \tilde{E}_{\text{loc}}^*]. \quad (13)$$

Inserting (6) and using $E_{\text{loc}} E_{\text{loc}}^* = |E_{\text{loc}}|^2 \in \mathbb{R}$ gives

$$\bar{P} = \frac{\omega}{2} \text{Im}[\alpha_m] |\tilde{E}_{\text{loc}}|^2. \quad (14)$$

The Meep source is not a single harmonic. Linearity of (6) means each Fourier component of E_{loc} drives its own $\mu(\omega)$ with its own $\alpha_m(\omega)$. The same bilinear form, evaluated on the DFTs (2), is the spectral density of the work (3). Thus,

$$\bar{P}(\omega) = \frac{\omega}{2} \text{Im}[\mu(\omega) E_{\text{loc}}^*(\omega)] = \frac{\omega}{2} \text{Im}[\alpha_m(\omega)] |E_{\text{loc}}(\omega)|^2. \quad (15)$$

3.3 Dissipated-power spectrum A_{diss}

Equation (15) is already the dissipated-power spectrum. To plot it next to a polarizability-like curve we give it the same $4\pi\omega/c$ dressing used later for a plane-wave cross section, and the same DFT sign flip that turns a physical absorption peak upward,

$$A_{\text{diss}}(\omega) = -\frac{4\pi\omega}{c} \text{Im}[\mu(\omega)E_{\text{loc}}^*(\omega)]. \quad (16)$$

Figure 6 is (16) for the hybrid job and for isolated Na. The hybrid curve has one molecular peak at 2.05 eV and a shoulder near 2.2 eV from $|E_{\text{loc}}(\omega)|^2$ (plasmon plus the Meep pulse).

Isolated A_{diss} uses the same formula, but there is no nanoparticle, so the field at the atom is the incident pulse, $\mu^{\text{iso}} = \alpha_m E_{\text{inc}}$. Equation (16) therefore collapses to

$$A_{\text{diss}}^{\text{iso}}(\omega) = -\frac{4\pi\omega}{c} \text{Im}[\mu^{\text{iso}}(\omega)E_{\text{inc}}^*(\omega)] = -\frac{4\pi\omega}{c} \text{Im}[\alpha_m(\omega)] |E_{\text{inc}}(\omega)|^2. \quad (17)$$

Because (17) still contains $|E_{\text{inc}}|^2$, peak-normalized isolated A_{diss} is not a Lorentzian: it follows the pulse envelope (center ~ 2.84 eV). The hybrid peak is ≈ 17.7 times the isolated peak, which is $|E_{\text{loc}}/E_{\text{inc}}|^2$ at 2.03 eV.

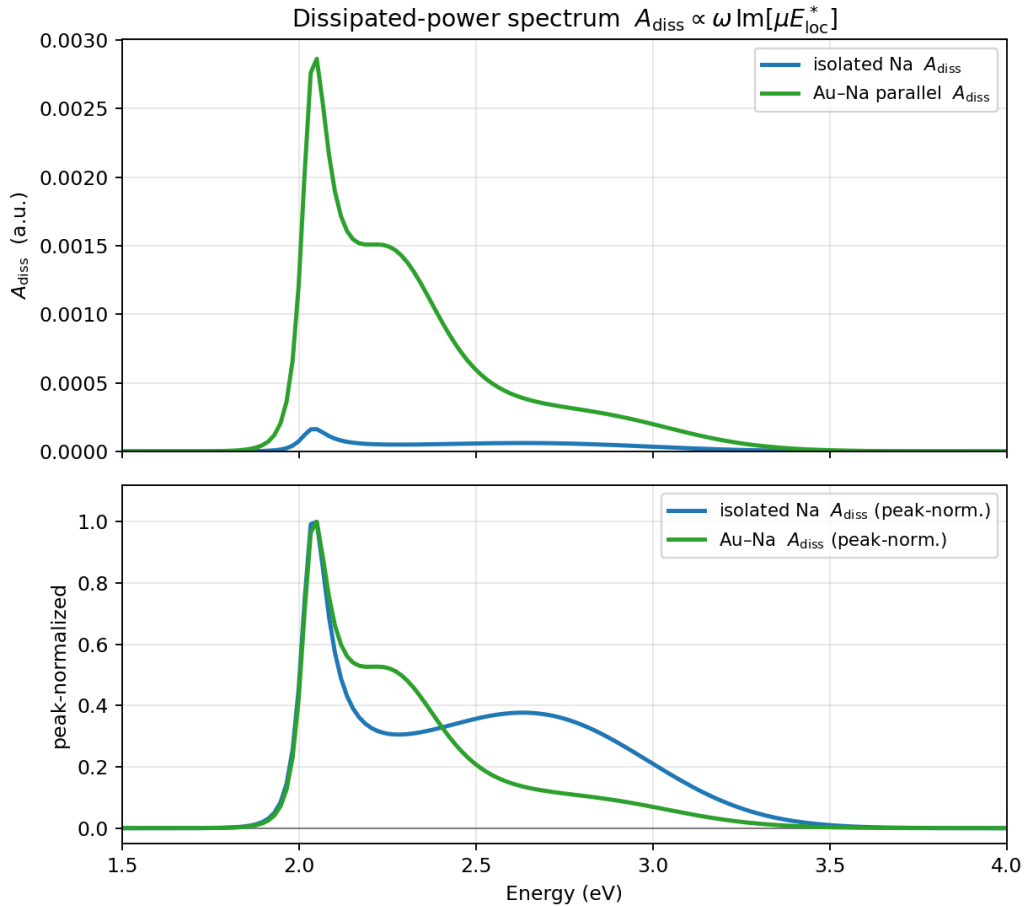


Figure 6: A_{diss} from (16) for Au–Na parallel (green) and from (17) for isolated Na (blue). Top: un-normalized. Bottom: peak-normalized. Isolated Na still carries $|E_{\text{inc}}|^2$; the hybrid extra height and the 2.2 eV shoulder come from $|E_{\text{loc}}|^2$.

In order to remove the peak shoulder seen in $A_{\text{diss}}^{\text{iso}}$, we should divide by the squared incident field, but doing so would give us a new quantity: molecular absorption cross section. That ratio is most cleanly written after introducing the local-field factor G .

3.4 Local field enhancement

The complex local-field factor is

$$G(\omega) := \frac{E_{\text{loc}}(\omega)}{E_{\text{inc}}(\omega)}. \quad (18)$$

Isolated Na would have $E_{\text{loc}}(\omega) = E_{\text{inc}}(\omega)$ so $G = 1$. The shared delay phase of the pulse drops out of $E_{\text{loc}}/E_{\text{inc}}$, so the structure left in G is the scattering from the NP.

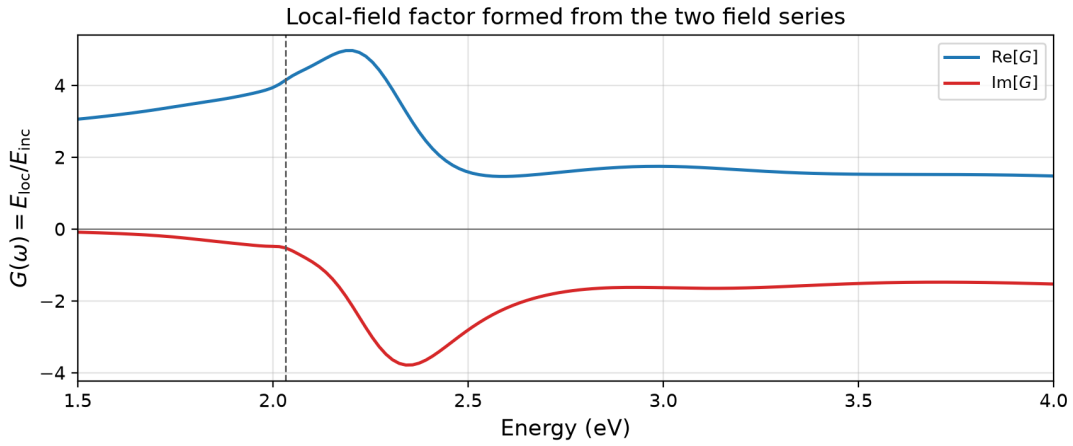


Figure 7: Local-field factor $G = E_{\text{loc}}/E_{\text{inc}}$. Dashed line: Na resonance.

We can then rewrite (6) in terms of $G(\omega)$:

$$\frac{\mu(\omega)}{E_{\text{inc}}(\omega)} = \alpha_m(\omega)G(\omega) \quad (19)$$

and if we define $\alpha_{\text{eff}}(\omega) := \alpha_m(\omega)G(\omega)$,

$$\mu(\omega) = \alpha_{\text{eff}}(\omega)E_{\text{inc}}(\omega). \quad (20)$$

So, α_{eff} is the molecular dipole expressed in the incident field. The molecule still responds to E_{loc} ; $\alpha_{\text{eff}} = \alpha_m G$ is that response as seen from the source. The extra height of hybrid A_{diss} is $|G|^2$: $|E_{\text{loc}}|^2 = |G|^2|E_{\text{inc}}|^2$, and $|G(2.03 \text{ eV})|^2 \approx 17.7$.

3.5 Molecular absorption cross section σ_{mol}

A cross section is power over incident flux. A plane wave in Gaussian units has time-averaged Poynting intensity [1]

$$I_{\text{inc}} = \frac{c}{8\pi} |\tilde{E}_{\text{inc}}|^2, \quad (21)$$

and, for the pulse, the same object built from the DFT of the vacuum reference,

$$I_{\text{inc}}(\omega) = \frac{c}{8\pi} |E_{\text{inc}}(\omega)|^2. \quad (22)$$

The molecular absorption cross section referred to the vacuum wave in (22) is

$$\sigma_{\text{mol}}(\omega) := \frac{\bar{P}(\omega)}{I_{\text{inc}}(\omega)} = \frac{4\pi\omega}{c} \frac{\text{Im}[\mu(\omega)E_{\text{loc}}^*(\omega)]}{|E_{\text{inc}}(\omega)|^2}. \quad (23)$$

With (6) and $E_{\text{loc}} = GE_{\text{inc}}$ this is

$$\sigma_{\text{mol}}(\omega) = \frac{4\pi\omega}{c} \text{Im}[\alpha_m(\omega)]|G(\omega)|^2. \quad (24)$$

The effect of the nanoparticle enters from $|G|^2$. A passive molecule (one with no significant emission back into the system) has $\text{Im}[\alpha_m] \geq 0$, so $\sigma_{\text{mol}} \geq 0$.

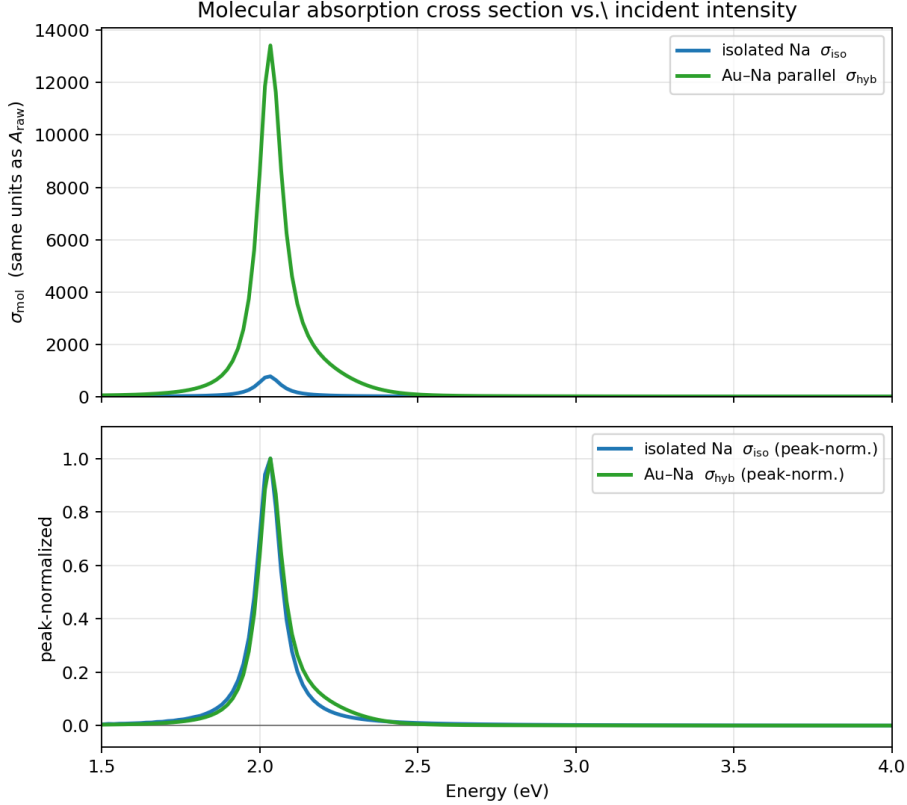


Figure 8: σ_{mol} for Au–Na parallel (green) and isolated Na (blue). Top: un-normalized. Bottom: peak-normalized.

3.6 Peak-normalized difference of the two cross sections

After peak-normalization the two curves in Figure 8 look almost the same, but they are not identical. Normalizing each spectrum to its own maximum strips the overall $|G|^2$ height (the hybrid peak is ≈ 17.7 times the isolated peak) so the residual lineshape mismatch is visible. Both maxima sit at 2.033 eV. The difference

$$\Delta(E) = \frac{\sigma_{\text{mol}}^{\text{hyb}}(E)}{\sigma_{\text{mol}}^{\text{hyb}, \text{max}}} - \frac{\sigma_{\text{mol}}^{\text{iso}}(E)}{\sigma_{\text{mol}}^{\text{iso}, \text{max}}} \quad (25)$$

is therefore zero at 2.033 eV by construction (Figure 9). The leftover is an odd swing across the Na line: $\min \Delta = -0.071$ at 1.999 eV and $\max \Delta = +0.069$ at 2.084 eV, from the slow rise of $|G(\omega)|^2$ across the resonance. The hybrid line is leaner on the red side and heavier on the blue;

past ~ 2.5 eV the peak-normalized spectra agree to less than 1%.

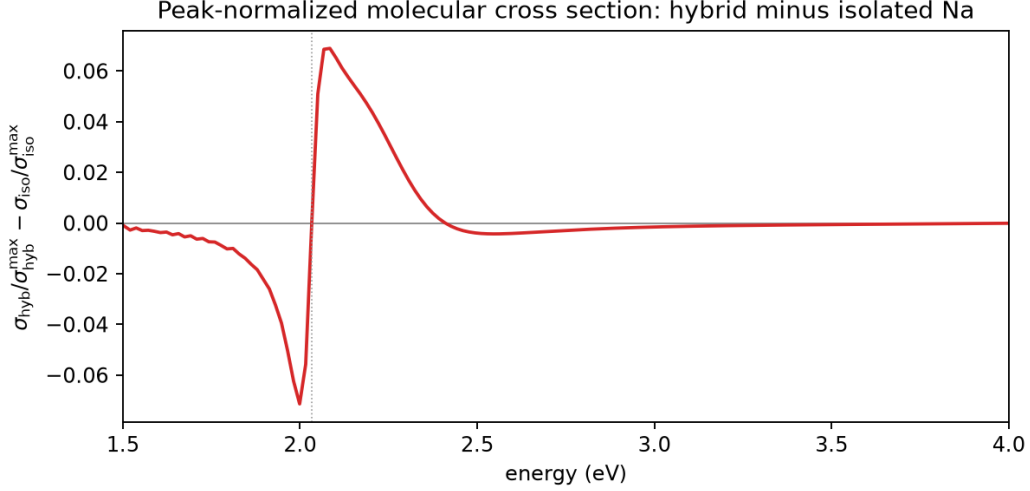


Figure 9: Peak-normalized hybrid σ_{mol} minus peak-normalized isolated Na. The dotted line is 2.033 eV. The odd residual is the slow tilt of $|G|^2$ across the molecular line, not a second resonance.

4 Modified power using E_{inc}

4.1 Why refer the same dipole to E_{inc}

Both A_{diss} and σ_{mol} are built from the field that actually sits on the atom, so they answer how much the molecule absorbs. They cannot tell us how that absorption appears to the incoming wave: E_{loc} already contains the nanoparticle, and dividing by it (or contracting with it) removes the phase of the gap field. The experimentally controlled quantity is the incident pulse E_{inc} .

E_{loc} tells us what the molecule is. E_{inc} tells us what the hybrid setup does to the light we sent. An optical experiment does not have a probe in the 1.45nm gap; it knows the wave that was launched. A Gersten–Nitzan model is written in that same field,

$$\alpha_{\text{eff}} = \frac{\alpha_m G}{1 - \alpha_m S} = \frac{\mu}{E_{\text{inc}}}. \quad (26)$$

If we divide μ by E_{loc} we undo the nanoparticle on purpose and recover α_m . That is a useful check (and it is how one confirms that back-action is weak: $|\alpha_m^{\text{hyb}}/\alpha_m^{\text{iso}}| \approx 0.99$ at 2.03eV). It is not a hybrid lineshape.

The same dipole $\mu(t)$, referred to E_{inc} instead of E_{loc} , is therefore the molecular response as seen from the source. We form it first as a bookkeeping analogue of (3),

$$P_{\text{inc}}(t) = \dot{\mu}(t) E_{\text{inc}}(t), \quad (27)$$

which is *not* the mechanical power into the atom. It is (3) with the partner field replaced by the vacuum wave.

$$\bar{P}_{\text{inc}}(\omega) = \frac{\omega}{2} \text{Im}[\mu(\omega) E_{\text{inc}}^*(\omega)]. \quad (28)$$

Dividing by the incident intensity (22) and using

$$\frac{\mu E_{\text{inc}}^*}{|E_{\text{inc}}|^2} = \frac{\mu}{E_{\text{inc}}} \quad (29)$$

gives

$$A_{\text{inc}}(\omega) := \frac{\overline{P}_{\text{inc}}(\omega)}{I_{\text{inc}}(\omega)} = \frac{4\pi\omega}{c} \text{Im}\left[\frac{\mu(\omega)}{E_{\text{inc}}(\omega)}\right] = \frac{4\pi\omega}{c} \text{Im}[\alpha_{\text{eff}}(\omega)]. \quad (30)$$

This is the vacuum dipole formula [1] with $\alpha \mapsto \alpha_{\text{eff}}$ and $E \mapsto E_{\text{inc}}$.

4.2 The driver spectrum A_{raw}

PlasMol plots (30) with a global minus so that a physical $3s \rightarrow 3p$ peak points up. With the DFT (2), $\text{Im}[\mu/E_{\text{inc}}]$ is negative at a passive Lorentzian, and

$$A_{\text{raw}}(\omega) = -\frac{4\pi\omega}{c} \text{Im}\left[\frac{\mu(\omega)}{E_{\text{inc}}(\omega)}\right] = -\frac{4\pi\omega}{c} \text{Im}[\alpha_{\text{eff}}(\omega)] \quad (31)$$

is the upright copy of A_{inc} . After peak-normalization, A_{raw} and A_{inc} are the same curve up to that sign. In the code, ω on the prefactor is the energy axis in eV while c is c_{au} , so the absolute scale is not a clean a.u.² cross section; only the shape is used.

Expanding the imaginary part,

$$\text{Im}[\alpha_{\text{eff}}] = \text{Im}[\alpha_m G] = \text{Im}[\alpha_m] \text{Re}[G] + \text{Re}[\alpha_m] \text{Im}[G]. \quad (32)$$

The first term is the even Lorentzian, weighted by $\text{Re}[G]$. The second is the odd (dispersive) wing of α_m weighted by the phase of the near field. A Lorentzian model (Figure 10) has $\text{Re}[\alpha_m] > 0$ below $\omega_m = 2.033\text{eV}$, 0 on resonance, and $\text{Re}[\alpha_m] < 0$ above it. As soon as $\text{Im}[G] \neq 0$, the second term goes negative on the blue side and can overcome the Lorentzian. That product is the lobe at 2.32eV in Figure 11. Isolated Na has $\text{Im}[G] \simeq 0$, so A_{raw} stays non-negative. The delay beating of Figure 3 cancels in the ratio μ/E_{inc} ; what remains is α_{eff} , whose imaginary part (sign-flipped) is A_{raw} .

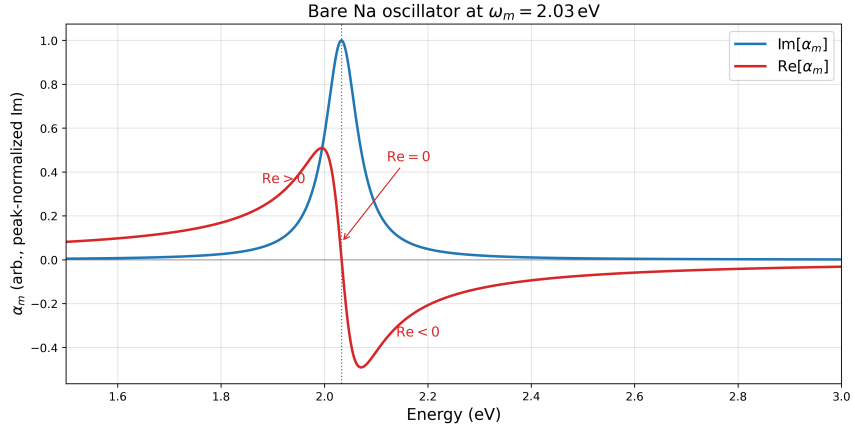


Figure 10: Model $\text{Re}[\alpha_m]$ (red, odd) and $\text{Im}[\alpha_m]$ (blue, even) about 2.033eV. The red wing is what $\text{Im}[G]$ multiplies in (32).

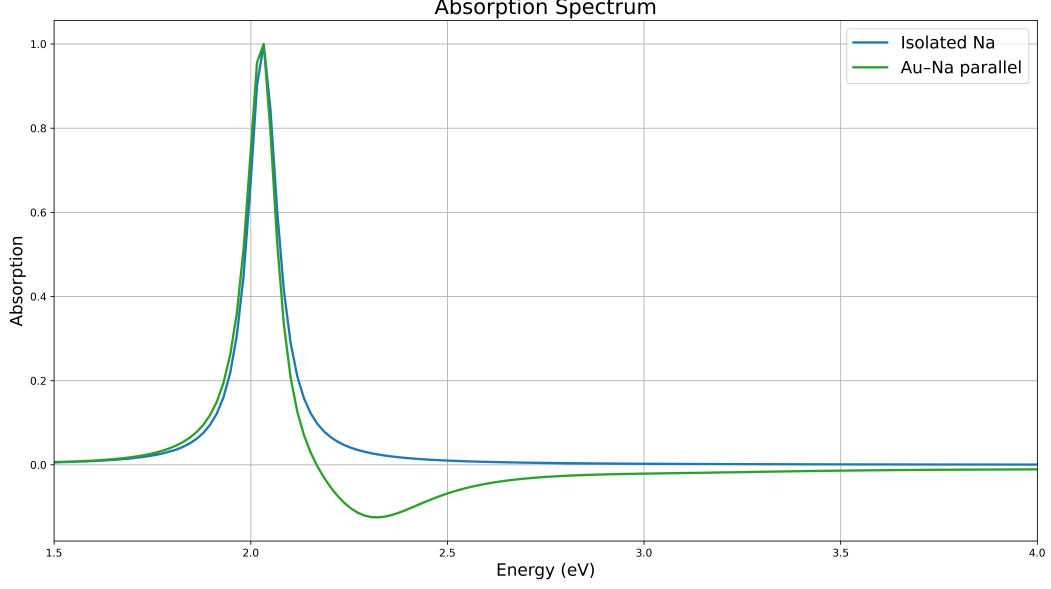


Figure 11: Peak-normalized A_{raw} for Au–Na parallel (green) and isolated Na (blue), 1.5–4.0eV. The hybrid curve crosses zero at 2.17eV and reaches ≈ -0.125 at 2.32eV.

μ in (31) is still only the atom. A_{raw} is not the extinction of Au+Na, and it is not $\text{Im}[\mathbf{p}_{\text{tot}}/E_{\text{inc}}]$. It is the molecular dipole expressed in the only field an experiment or a classical hybrid model actually knows.

5 The three observables together

The same three time series produce three different spectra, in the order they were derived (Figure 12),

$$A_{\text{diss}}(\omega) = -\frac{4\pi\omega}{c} \text{Im}[\mu(\omega)E_{\text{loc}}^*(\omega)], \quad (33)$$

$$\sigma_{\text{mol}}(\omega) = \frac{4\pi\omega}{c} \frac{\text{Im}[\mu(\omega)E_{\text{loc}}^*(\omega)]}{|E_{\text{inc}}(\omega)|^2} = \frac{4\pi\omega}{c} \text{Im}[\alpha_m(\omega)]|G(\omega)|^2, \quad (34)$$

$$A_{\text{raw}}(\omega) = -\frac{4\pi\omega}{c} \text{Im}\left[\frac{\mu(\omega)}{E_{\text{inc}}(\omega)}\right]. \quad (35)$$

They are not three normalizations of one curve. The split is which field sits next to μ , and whether one then divides by I_{inc} .

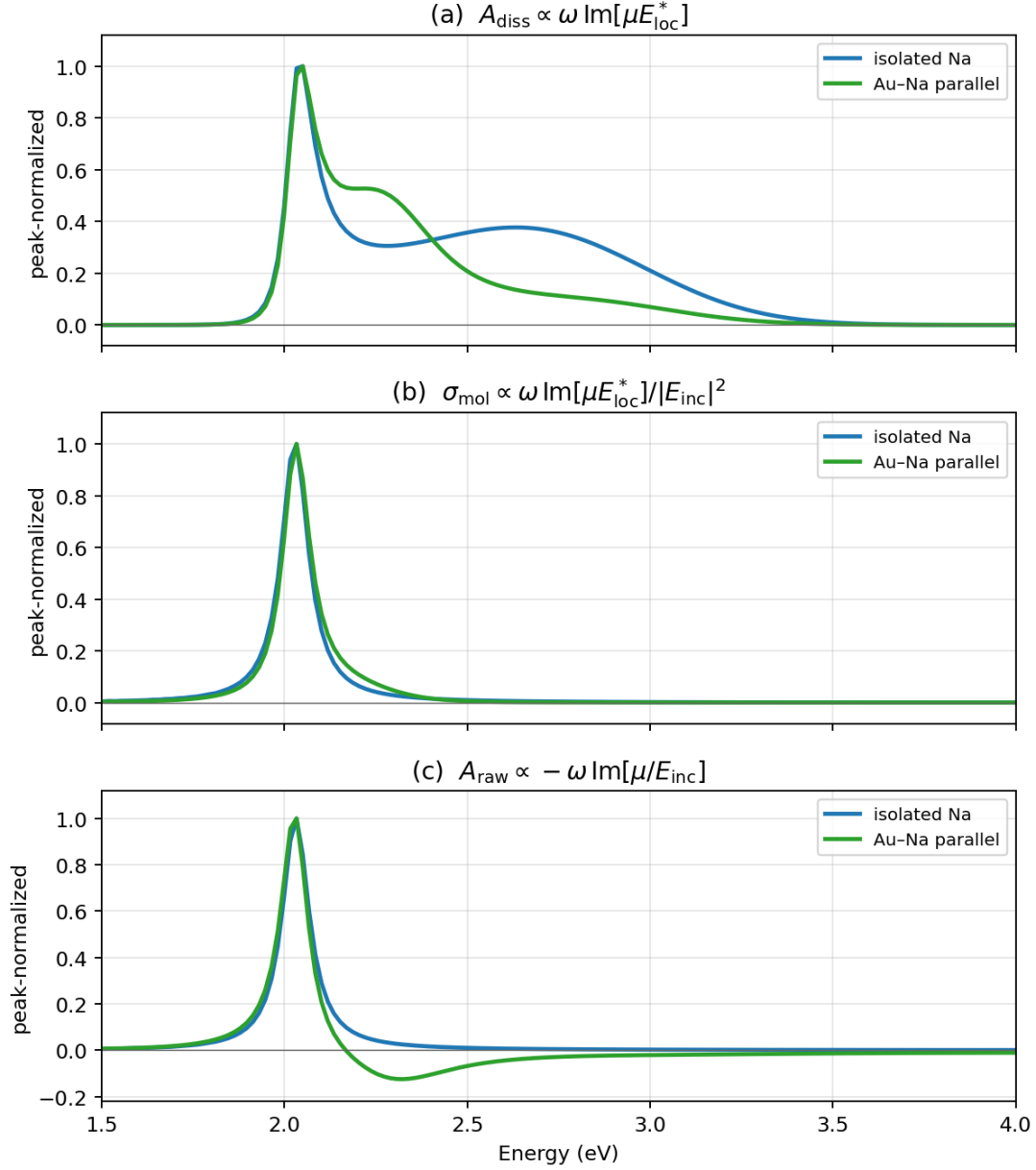


Figure 12: Peak-normalized hybrid (green) and isolated-Na (blue) spectra, in the order of Sections 3.3–4.2. (a) A_{diss} , still carrying $|E_{\text{inc}}|^2$ when isolated. (b) σ_{mol} , the pulse divided out. (c) A_{raw} ; this is the only panel allowed to go negative.

Acknowledgements of convention

Phasors follow $e^{-i\omega t}$ [1]. The cycle average $\frac{1}{2} \text{Re}[\tilde{A}\tilde{B}^*]$ is Jackson §6.9; the vacuum intensity $c|E|^2/8\pi$ is the Gaussian form of §7.1. The step from those two formulae to $\sigma = (4\pi\omega/c) \text{Im}[\alpha]$ is the dipole reduction of the optical theorem, not a single numbered equation in Jackson. The minus signs in (16) and (31) are DFT bookkeeping, not additional physics.

References

- [1] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. (Wiley, 1999), §6.9 and §7.1.

- [2] L. Novotny and B. Hecht, *Principles of Nano-Optics*, 2nd ed. (Cambridge University Press, 2012).